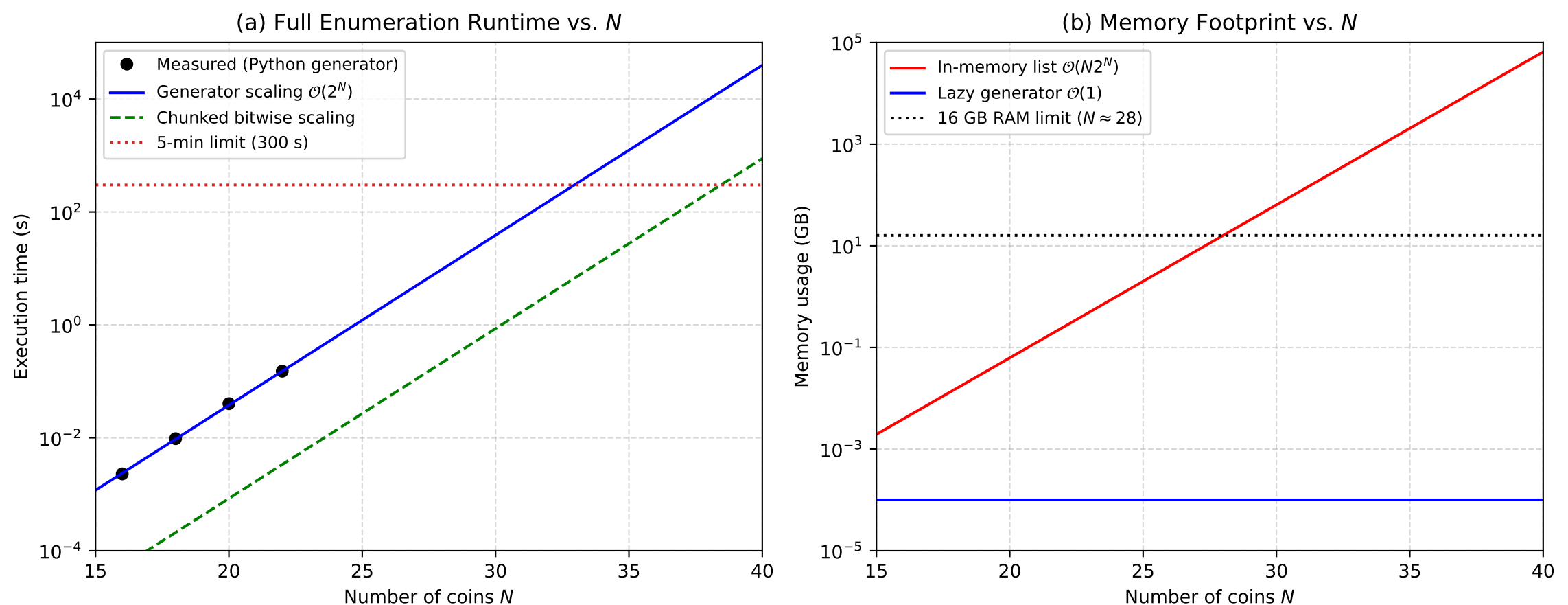
(a) Full Enumeration Runtime vs. N
Measured (Python generator)
Generator scaling $\mathcal{O}(2^N)$
Chunked bitwise scaling
5-min limit (300 s)
Execution time (s)
Number of coins N
(b) Memory Footprint vs. N
In-memory list $\mathcal{O}(N2^N)$
Lazy generator $\mathcal{O}(1)$
16 GB RAM limit ($N \approx 28$)
Memory usage (GB)
Number of coins N